Class #10 - Orthogonality

1. Def: Let u and v be vectors in $\mathbf{R^n}$.
   The product $u^T v$ is called the *scalar product* or _____of u and v (It is often written as $u \cdot v$).

2. Theorem: Let u, v, and w be vectors in $\mathbf{R^n}$, and let c be a scalar. Then

   (a) $u \cdot v =$ _____

   (b) $(u + v) \cdot w =$ _____

   (c) $(cu) \cdot v =$ _____ = _____

   (d) $u \cdot u \geq$ _____, and $u \cdot u = 0$ if and only if _____.

3. Def: The *length* (or *norm*) of v is the _____scalar $||v||$ defined by $||v||$ = _____ = _____, and $||v||^2 =$ _____.
   Note: $||cv||$ = _____ for any scalar c.

4. Def: For u and v in $\mathbf{R^n}$, the *distance between u and v*, written as $\text{dist}(u, v)$, is the length of the vector $u - v$. That is,
$$\text{dist}(u, v) = ||u - v||.$$

5. For two *nonzero* vectors u and v in $\mathbf{R^2}$ or in $\mathbf{R^3}$,
$$u \cdot v = _____$$
   where θ is the angle between u and v, _____.

6. Two *nonzero* vectors are _____if the angle between them is $\theta = \pi/2$. $\Rightarrow$ _____ $= 0$.
   We generalize this fact for any vectors (even for zero vectors) in $\mathbf{R^n}$.

7. Def: two vectors u and v in $\mathbf{R^n}$ are *orthogonal* (to each other) if $u \cdot v = 0$.
   Note: _____is orthogonal to every vector in $\mathbf{R^n}$.

8. The Pythagorean Theorem:
   Two vectors u and v are orthogonal if and only if $||u+v||^2 =$ _____ + _____.
   Proof:

9. Def: Let W be a subspace of $\mathbf{R^n}$. If a vector z is orthogonal to every vector in W, then z is said to be *orthogonal to* W.
   The set of all vecters z that are orthogonal to W is called the *orthogonal complement of* W and is denoted by $W^{\perp}$.

10. Examples: Suppose $n \geq 2$.
    If $W = \{0\}$, then $W^{\perp} =$ _____
    If $\dim W = 1$, then $W =$ _____, and there exists a nonzero vector x in $\mathbf{R^n}$ such that _____.
    Then x is orthogonal to _____of W.

Fill in the blank: A vector $x \in W^{\perp}$ _____x is orthogonal to every vector in a set that spans W.

If dim $W = n$, then $W^{\perp} =$ _____

11. Theorem: Let W be a subspace of $\mathbf{R^n}$. Then $W^{\perp}$ is a _____of $\mathbf{R^n}$.
    Proof:

12. Example: Let A be an $m \times n$ matrix. Then, $(\text{Nul}A)^{\perp} =$ _____.